

8.4 Testing and Validation Status

Testing is being conducted in parallel with development. The following testing phases have been executed:

1. Unit Testing:

- Backend: Jest is used to test individual API endpoints. Currently, we have achieved 75% code coverage for the Booking Service.
- ML Models: We have validated the models against a hold-out test set (20% of data), achieving the targeted 94% accuracy for the Random Forest classifier.

2. Integration Testing:

- We have successfully verified the data flow from App -> API Gateway -> Database.
- We have verified the handshake between the Node.js backend and the Python ML service.

8.5 Current Challenges and Mitigation Strategies

While the project is largely on track, we have encountered and are addressing the following challenges:

- **Challenge:** Latency in ML Predictions. Initial API calls to the Python container took ~3 seconds due to model loading times.
 - *Mitigation:* We refactored the code to load the models into memory only once during server startup (as seen in Code Snippet 2), reducing inference time to <200ms.
- **Challenge:** Map API Costs. During testing, frequent calls to the Google Maps API for auto-complete addresses consumed quota rapidly.
 - *Mitigation:* We implemented 'debounce' logic in the frontend to only send API requests after the user stops typing for 500ms.

8.6 Conclusion of Status

The SHIFTY project has successfully moved past the proof-of-concept stage. The robust integration of the MERN stack (MongoDB, Express, React Native, Node) with Python-based AI services demonstrates a high level of technical maturity.

The remaining 30% of the project timeline is dedicated to **User Acceptance Testing (UAT)** and **System Hardening**. We are confident that the application will be feature-complete and ready for final presentation by the scheduled deadline.